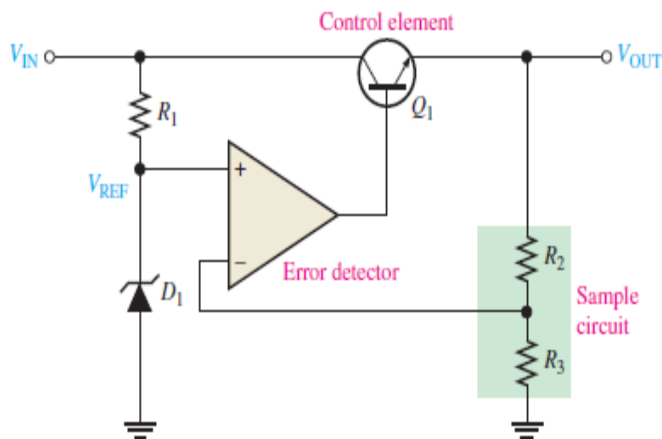


TUTORIAL 02

1. Explain the concept of cutoff frequency in the context of electronic filters.
2. Discuss the significance of filter order in electronic filter design.
3. How are the Q- factor and the bandwidth of a band-pass filter related? Explain how the selectivity is affected by the Q-factor of a filter.
4. What is a Butterworth filter?
5. Explain the difference between a series linear regulator and a shunt linear regulator.
6. Calculate the power dissipation in a linear regulator if the input voltage is 12V, the output voltage is 5V, and the load current is 1A.
7. A certain voltage regulator has a 12 V output when there is no load ($I_L=0$). When there is a full-load current of 10 mA, the output voltage is 11.9 V. Express the voltage regulation as a percentage change from no-load to full-load and also as a percentage change for each mA change in load current.
8. Explain the regulating action of Basic op-amp series regulator circuit below when V_{IN} or R_L connected at the output either:
 - a) Increases
 - b) Decreases



9. What are the basic components in a series regulator?
10. What is the primary advantage of switching regulators over linear regulators?
11. Design a linear voltage regulator circuit to provide a stable 5V output from a 12V input supply using a Zener diode and an NPN transistor. The load current ranges from 0 to 100 mA. Choose appropriate values for the Zener diode, the base resistor for the transistor, and the series resistor for the Zener diode. Explain your choices and show your calculations.